

## Lecture 1

"Infinitely Often" (i.o.) Events

Lemma

Borel - Cantelli

Probability Converge

### "Infinitely Often" (i.o.) Events

312 e.g

Flip infinitely sequence of fair indep coins

$$H_n = \{n^{\text{th}} \text{ coin Heads}\} \rightarrow P(H_n) = \frac{1}{2}$$

What's  $P(H_n \text{ i.o.}) = P(\text{infinitely many of the coins are heads})$ ?

Or e.g. suppose  $\{B_n\}$  indep,  $P(B_n) = \frac{1}{n}$   $P(B_n \text{ i.o.}) = ?$

e.g. suppose  $\{C_n\}$  indep,  $P(C_n) = \frac{1}{n^2}$   $P(C_n \text{ i.o.}) = ?$

Defn: Given events  $A_1, \dots, A_n, \dots$ , define

$$\{A_n \text{ i.o.}\} = \{ \forall n, \exists k \geq n \text{ s.t. } A_k \text{ occurs} \}$$

$$= \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \limsup_n A_n$$

为了有方法去算  $P(A_n \text{ i.o.})$

① Lemma  $P(A_n \text{ i.o.}) \geq \limsup_{n \rightarrow \infty} P(A_n)$

② Borel - Cantelli Lemma

① Lemma:

$$P(A_n \text{ i.o.}) \geq \limsup_{n \rightarrow \infty} P(A_n)$$

Proof:

$$P(A_n \text{ i.o.}) = P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = P\left(\bigcap_{n=1}^{\infty} B_n\right) \quad \leftarrow B_n = \bigcup_{k=n}^{\infty} A_k$$

then, we find out

$$\begin{aligned} B_1 &= A_1 \cup A_2 \cup \dots \\ B_2 &= A_2 \cup A_3 \cup \dots \\ B_n &= A_n \cup A_{n+1} \cup \dots \\ B_{n+1} &= A_{n+1} \cup A_{n+2} \cup \dots \end{aligned}$$

$$\Rightarrow B_1 \supseteq B_2 \supseteq B_3 \supseteq \dots \supseteq B_n \supseteq B_{n+1} \supseteq \dots$$

i.e.  $\{B_n\} \searrow B$

nested decreasing

Tip  $\{B_n\} \searrow B$

$$\Leftrightarrow \begin{cases} B_n \supseteq B_{n+1} & \forall n \\ \bigcap_{n=1}^{\infty} B_n = B \end{cases}$$

So, by continuity of probability,

$$\lim_{n \rightarrow \infty} P(B_n) = P\left(\bigcap_{n=1}^{\infty} B_n\right)$$

Tip 对于 Continuity of Probability 的证明

① Continuity from below

suppose  $A_1 \subseteq A_2 \subseteq \dots$

$$\text{WTS } \lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcup_{n=1}^{\infty} A_n\right)$$

Pf:

$$\text{Let } C_1 = A_1, C_2 = A_2 \setminus A_1, C_3 = A_3 \setminus A_2, \dots, C_n = A_n \setminus A_{n-1}$$

then  $C_i \cap C_j = \emptyset \quad \forall i \neq j$   $\leftarrow$  will be proved later

Then,

$$A_n = \bigcup_{k=1}^n C_k \quad \text{and} \quad P(A_n) = \sum_{k=1}^n P(C_k) = P\left(\bigcup_{k=1}^n C_k\right)$$

$$\lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^n P(C_k) = \sum_{k=1}^{\infty} P(C_k) = P\left(\bigcup_{k=1}^{\infty} C_k\right) = P\left(\bigcup_{n=1}^{\infty} A_n\right) \quad \square$$

Countable Additivity

Tip 为什么需要转这么一圈?

$$\lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} P\left(\bigcup_{k=1}^n C_k\right) \stackrel{??}{=} P\left(\bigcup_{k=1}^{\infty} C_k\right)$$

这层关系不知道

但是可以变成

$$\lim_{n \rightarrow \infty} P\left(\bigcup_{k=1}^n C_k\right) = \lim_{n \rightarrow \infty} \sum_{k=1}^n P(C_k) = \sum_{k=1}^{\infty} P(C_k) = P\left(\bigcup_{k=1}^{\infty} C_k\right)$$

Tip  $C_i \cap C_j = \emptyset \quad \forall i \neq j$  证明

{ Nested Sequence  $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$   
 $C_1 = A_1 \quad C_n = A_n \setminus A_{n-1}$

假设  $i < j$ ,  $A_i \subseteq A_{j-1}$

$$C_i = A_i \setminus A_{i-1} \Rightarrow C_i \subseteq A_i \Rightarrow C_i \subseteq A_{j-1}$$

$$C_j = A_j \setminus A_{j-1} \Rightarrow C_j = A_j \cap (A_{j-1})^c \Rightarrow C_j \subseteq (A_{j-1})^c$$

$$\therefore C_i \cap C_j \subseteq A_{j-1} \cap (A_{j-1})^c = \emptyset$$

Tip 对于 Continuity of Probability 的证明

② Continuity from above

suppose  $B_1 \supseteq B_2 \supseteq \dots$

$$\text{WTS } \lim_{n \rightarrow \infty} P(B_n) = P\left(\bigcap_{n=1}^{\infty} B_n\right)$$

Pf:  $B_1 \supseteq B_2 \supseteq \dots \Rightarrow B_1^c \subseteq B_2^c \subseteq \dots$

Applying continuity from below to  $\{B_n^c\}$ ,

$$\lim_{n \rightarrow \infty} P(B_n^c) = P\left(\bigcup_{n=1}^{\infty} B_n^c\right) \xleftarrow{\text{De Morgan}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} [1 - P(B_n)] = P\left[\left(\bigcap_{n=1}^{\infty} B_n\right)^c\right] = 1 - P\left(\bigcap_{n=1}^{\infty} B_n\right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} P(B_n) = P\left(\bigcap_{n=1}^{\infty} B_n\right)$$

$$\text{Finally, } P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) = P\left(\bigcap_{n=1}^{\infty} B_n\right) \underset{\substack{\uparrow \\ \text{decreasing}}}{=} \lim_{n \rightarrow \infty} P(B_n) = \limsup_{n \rightarrow \infty} P\left(\bigcup_{k=n}^{\infty} A_k\right) \geq \limsup_{n \rightarrow \infty} P(A_n)$$

无 Tip 版本

# ① Lemma

$$P(A_n \text{ i.o.}) \geq \limsup_{n \rightarrow \infty} P(A_n)$$

Proof:

$$P(A_n \text{ i.o.}) = \lim_{n \rightarrow \infty} P\left(\bigcap_{k=n}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) \quad B_{n+1} \subseteq B_n \subseteq B_{n-1}$$

i.e.  $\{B_n\} \searrow B$

$$\lim_{n \rightarrow \infty} P(B_n) = P\left(\bigcap_{n=1}^{\infty} B_n\right)$$

By Continuity of Probability

$$P(A_n \text{ i.o.})$$

$$= \lim_{n \rightarrow \infty} P\left(\bigcup_{k=n}^{\infty} A_n\right) \left[ \geq \lim_{n \rightarrow \infty} P(A_n) \quad \begin{array}{l} \text{不一定成立, 因为 } P(A_n), A_n \\ \text{不一定 converge} \end{array} \right]$$

$$= \limsup_{n \rightarrow \infty} P\left(\bigcup_{k=n}^{\infty} A_n\right)$$

$$\geq \limsup_{n \rightarrow \infty} P(A_n) \leftarrow \left[ \begin{array}{l} A_n \subseteq \bigcup_{k=n}^{\infty} A_n \Rightarrow P(A_n) \leq P\left(\bigcup_{k=n}^{\infty} A_n\right) \\ \text{且 } \limsup_{n \rightarrow \infty} P(A_n) \text{ 存在} \\ \text{since } 0 \leq P(A_n) \leq 1 \end{array} \right]$$

□

回到开头的例子

e.g. ① Coins:  $P(H_n \text{ i.o.}) \geq \limsup_{n \rightarrow \infty} \left(\frac{1}{2}\right) = \frac{1}{2}$

②  $\{B_n\}$ :  $P(B_n \text{ i.o.}) \geq \limsup_{n \rightarrow \infty} P\left(\frac{1}{n}\right) = 0$

$\uparrow$   
 $P(B_n) = \frac{1}{n}$

出现问题, 没有提供信息

## ② Borel - Cantelli Lemma

If  $A_1, A_2, \dots$  are events, then

(i) If  $\sum_{n=1}^{\infty} P(A_n) < \infty$ , then  $P(A_n \text{ i.o.}) = 0$

(ii) If  $\sum_{n=1}^{\infty} P(A_n) = \infty$ , and  $\{A_n\}$  independent, then  $P(A_n \text{ i.o.}) = 1$

Proof : (i)

Let  $m \in \mathbb{N}$ , Then

$$P(A_n \text{ i.o.}) = P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k\right) \leq P\left(\bigcup_{k=m}^{\infty} A_k\right) \\ \leq \sum_{k=m}^{\infty} P(A_k)$$

$$\begin{aligned} \text{Then, } 0 \leq P(A_n \text{ i.o.}) &\leq \sum_{k=m}^{\infty} P(A_k) \xrightarrow{\text{as } m \rightarrow \infty} 0 \quad \text{since } \sum_{n=1}^{\infty} P(A_n) < \infty \\ &\uparrow \\ &\text{Does not depend on } m, \\ &\text{所以 } m \rightarrow \infty \text{ 不影响.} \end{aligned}$$

Proof : (ii)

Key

证明思路

$$P(A_n \text{ i.o.}) = 1 \Leftrightarrow P(\{A_n \text{ i.o.}\}^c) = 0$$

$$\Leftrightarrow P\left(\bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c\right) = 0 \Leftrightarrow P\left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c\right)$$

$$\leq \sum_{n=1}^{\infty} P\left(\bigcap_{k=n}^{\infty} A_k^c\right) = 0$$

$$\therefore \text{WTS } P\left(\bigcap_{k=n}^{\infty} A_k^c\right) = 0 \quad \forall n$$

Tip

$$1+x \leq e^x \quad \forall x$$

$$\text{Proof: } f(x) = e^x - (1+x)$$

$$f'(x) = e^x - 1 = 0 \Rightarrow x = 0$$

$$f''(x) = e^x > 0 \quad \forall x \in \mathbb{R}$$

$$\begin{aligned} &\text{Graph of } f(x) = e^x - (1+x) \\ &f(0) = 1 - 1 = 0 \\ &\Rightarrow e^x - (1+x) \geq f(0) = 0 \end{aligned}$$

Continue Proof: (ii)

Let  $n \in \mathbb{N}, m \in \mathbb{N}$ .

Then,

$$\{ \bigcap_{k=n}^{\infty} A_k^c \} \subseteq \{ \bigcap_{k=n}^{n+m} A_k^c \}$$

$$P(\bigcap_{k=n}^{\infty} A_k^c) \leq P(\bigcap_{k=n}^{n+m} A_k^c)$$

Independent

$$= \prod_{k=n}^{n+m} P(A_k^c)$$

$$= \prod_{k=n}^{n+m} [1 - P(A_k)]$$

$$\leq \prod_{k=n}^{n+m} e^{-P(A_k)} = e^{-\sum_{k=n}^{n+m} P(A_k)}$$

$$= e^{-\sum_{k=n}^{\infty} P(A_k)}$$

Tip

独立

$$\Leftrightarrow P(A \cap B) = P(A)P(B)$$

Disjoint

$$\Leftrightarrow P(A \cap B) = 0$$

$\therefore$  当 Disjoint 时,

$$P(A \cup B) = P(A) + P(B)$$

$\therefore$  We have the inequality that  $\forall n$ ,

$$0 \leq P(\bigcap_{k=n}^{\infty} A_k^c) \leq e^{-\sum_{k=n}^{\infty} P(A_k)} \quad (\forall m)$$

$\downarrow$  when  $m \rightarrow \infty$

$$\text{So, } P(\bigcap_{k=n}^{\infty} A_k^c) = 0 \quad (\forall n)$$

Finally,

$$P(A_n \text{ i.o.}) = 1 - P(\{A_n \text{ i.o.}\}^c)$$

$$= 1 - P(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k^c)$$

$$\geq 1 - \sum_{n=1}^{\infty} P(\bigcap_{k=n}^{\infty} A_k^c) = 1$$

$$\therefore P(A_n \text{ i.o.}) = 1$$

□

回到最开始的例子

Coins:  $\{H_n\}$  indep 且  $\sum_{n=1}^{\infty} P(H_n) = \sum_{n=1}^{\infty} \frac{1}{2} = \infty$

$$\longrightarrow P(H_n \text{ i.o.}) = 1$$

$\{B_n\}$ :  $\{B_n\}$  indep 且  $\sum_{n=1}^{\infty} P(B_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$

$$\longrightarrow P(B_n \text{ i.o.}) = 1$$

$\{C_n\}$ :  $\sum_{n=1}^{\infty} P(C_n) = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$

$$\longrightarrow P(C_n \text{ i.o.}) = 0$$

有没有  $P(A_n \text{ i.o.})$  不是 0 或 1 的例子

e.g. Flip inf. many fair indep coins

Set  $A_1 = A_2 = \dots = H_1 = \{\text{First coin Heads}\}$

$$\text{then, } P(A_n \text{ i.o.}) = P(H_1) = \frac{1}{2}$$

注意: 尽管  $\sum_{n=1}^{\infty} P(A_n) = \infty$ , 但  $\{A_n\}$  是 dependent,  
用不了 Borel - Cantelli.

一些可以算但是稍微复杂的例子

e.g.  $A_n = H_{2n-1} \cap H_{2n} = \{\text{coins } 2n-1 \text{ and coins } 2n \text{ are both heads}\}$

$$P(A_n) = P(H_{2n-1} \cap H_{2n}) \underset{\text{indep}}{=} P(H_{2n-1})P(H_{2n}) = \frac{1}{4}$$

Also  $\{A_n\}$  indep  $\leftarrow A_1 = H_1 \cap H_2 \quad A_2 = H_3 \cap H_4 \quad \dots$

$$\therefore P(A_n \text{ i.o.}) = 1$$

e.g.  $B_n = H_n \cap H_{n+1} = \{\text{Coins } n \text{ and } n+1 \text{ are heads}\}$

问题:  $B_1 = H_1 \cap H_2$   $B_2 = H_2 \cap H_3$  ----  
 $\{B_n\}$  dependent

解决方案:

构造  $\{B_{2m}\}_{m \geq 1} \leftrightarrow B_2, B_4, \dots$

$B_2 = H_2 \cap H_3$   $B_4 = H_4 \cap H_5$  ----

$\{B_{2m}\}$  independent  $+ \sum_{m=1}^{\infty} P(B_{2m}) = \sum_{m=1}^{\infty} \frac{1}{4} = \infty$ ,

by Borel-Cantelli,

$$P(B_{2m} \text{ i.o.}) = 1$$

$$\{B_{2m} \text{ i.o.}\} \subseteq \{B_n \text{ i.o.}\}$$

$$\Rightarrow P(B_n \text{ i.o.}) \geq P(B_{2m} \text{ i.o.}) = 1$$

## Convergence of Random Variables

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If flip  $n$  fair indep coins, then for large  $n$ ,  
fraction of Heads will probably ~~goes~~ close to  $\frac{1}{2}$ .

As  $n \rightarrow \infty$ , fraction  $\rightarrow \frac{1}{2}$ .

↑  
random

How to say in math?

我们只能要求概率  $\rightarrow 1$ , 因为事件总是随机的。

Defn: A sequence  $X_1, X_2, \dots, X_n$  of random variables converge in probability  
to a r.v  $\gamma$  (or constant)

iff

$$\forall \epsilon > 0, \lim_{n \rightarrow \infty} P(|X_n - \gamma| \geq \epsilon) = 0$$

(write  $X_n \xrightarrow{P} \gamma$ )

e.g.  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bern}(\frac{1}{n})$ , i.e.  $\begin{cases} P(X_n = 1) = \frac{1}{n} \\ P(X_n = 0) = 1 - \frac{1}{n} \end{cases}$

Then  $\forall \epsilon > 0$ ,

$$\{X_n \geq \epsilon\} \Rightarrow \{X_n \neq 0\}$$

$$P(|X_n - 0| \geq \epsilon) = P(X_n \geq \epsilon) \leq P(X_n \neq 0)$$

$$= P(X_n = 1) = \frac{1}{n} \rightarrow 0$$

$$\therefore X_n \xrightarrow{P} 0$$

e.g.  $U \sim \text{uniform}[0, 1]$   $X_n = 1_{U \leq \frac{1}{2} + \frac{1}{2n}}$   $\gamma = 1_{U \leq \frac{1}{2}}$

Then,  $\forall \epsilon > 0$ ,

$$P(|X_n - \gamma| \geq \epsilon) \leq P(X_n \neq \gamma) = P\left(\frac{1}{2} \leq U \leq \frac{1}{2} + \frac{1}{2n}\right)$$

$$= \frac{1}{2n} \rightarrow 0$$

$$\therefore X_n \xrightarrow{P} \gamma$$